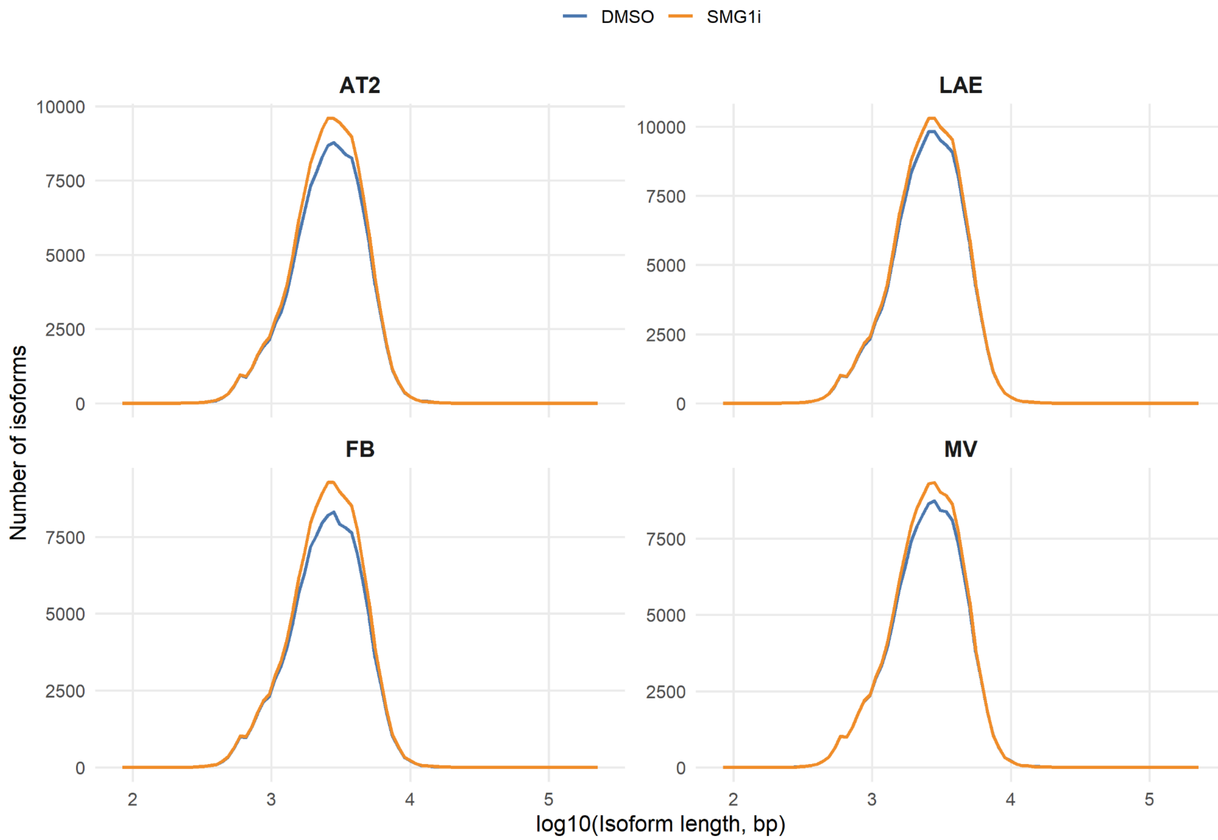
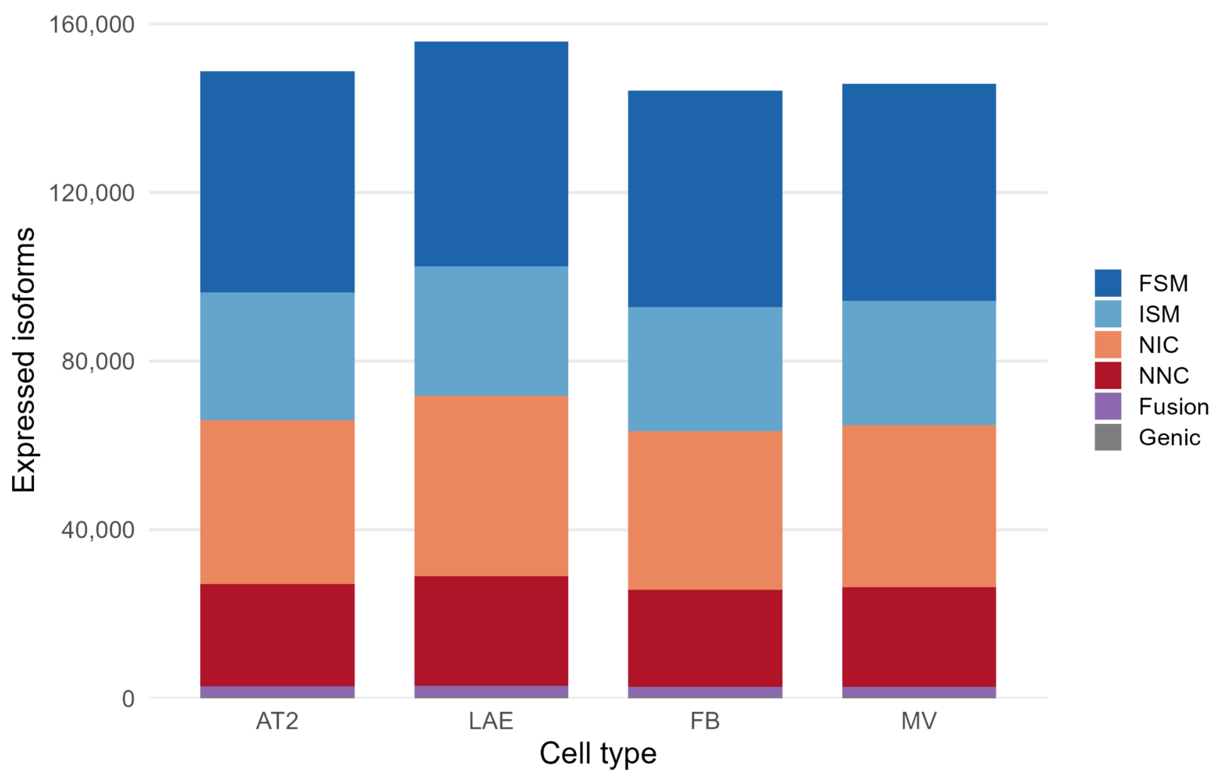
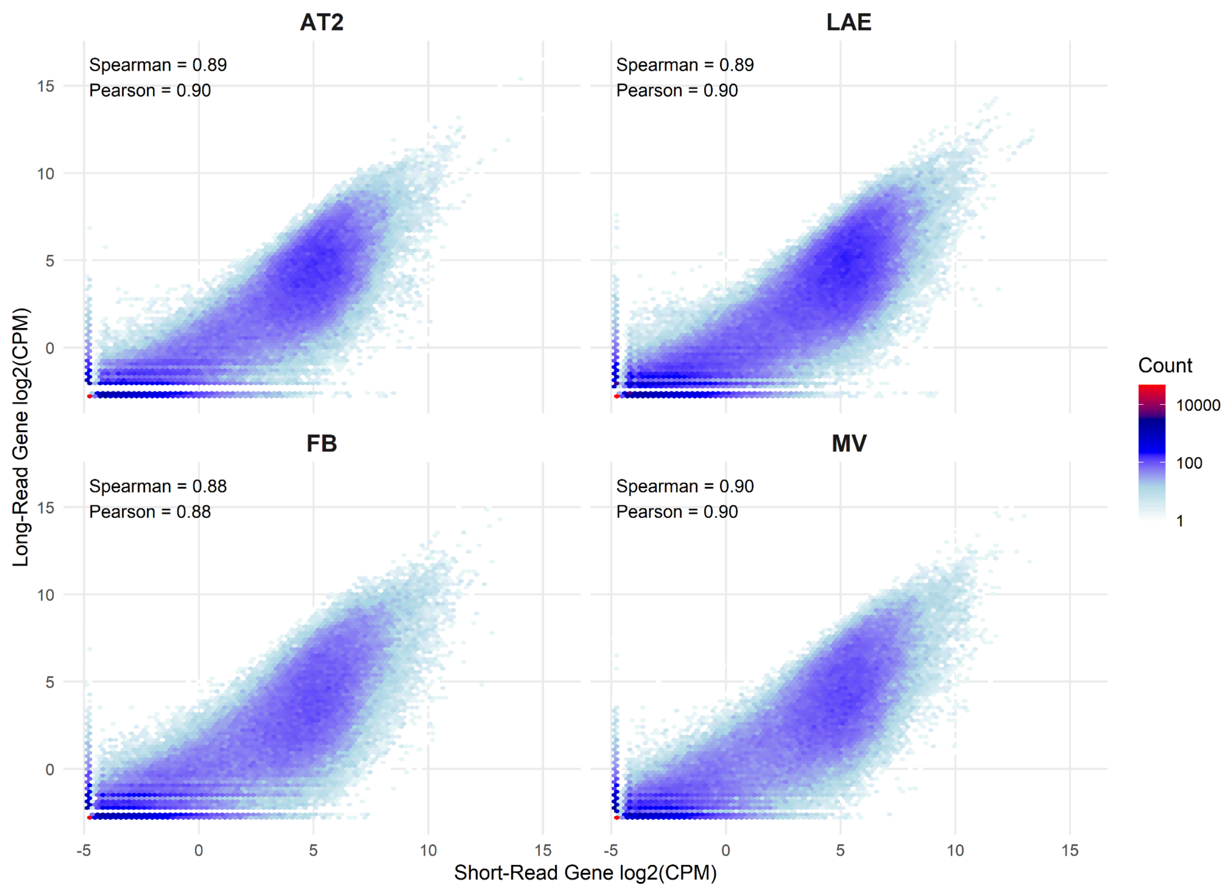
Manuscript Supplemental Figures



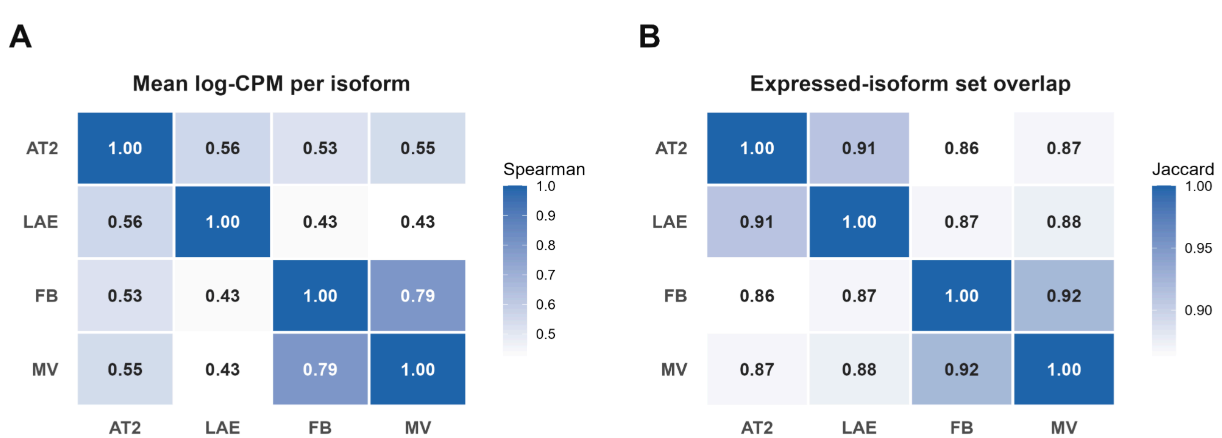
SF1 | Isoform Length by Cell Type and Treatment. Distribution of assembled long-read isoform lengths (log10 base pairs), shown as frequency curves for each of the four cell types (AT2, LAE, FB, MV) and overlaid by treatment (DMSO, SMG1i).



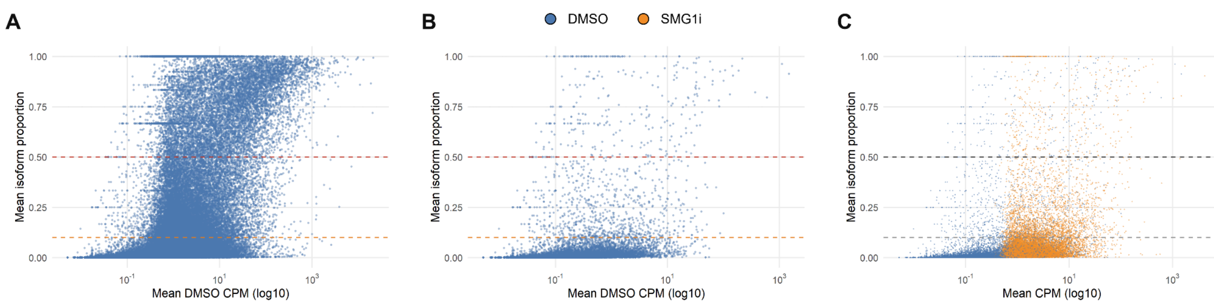
SF2 | SQANTI3 Structural Categories by Cell Type. Stacked bars of the number of expressed long-read isoforms in DMSO in each cell type (AT2, LAE, FB, MV), partitioned by SQANTI3 structural category (FSM, ISM, NIC, NNC, fusion, genic, antisense, intergenic, and genic-intron). **Abbreviations.** FSM, full splice match; ISM, incomplete splice match; NIC, novel in catalog; NNC, novel not in catalog.



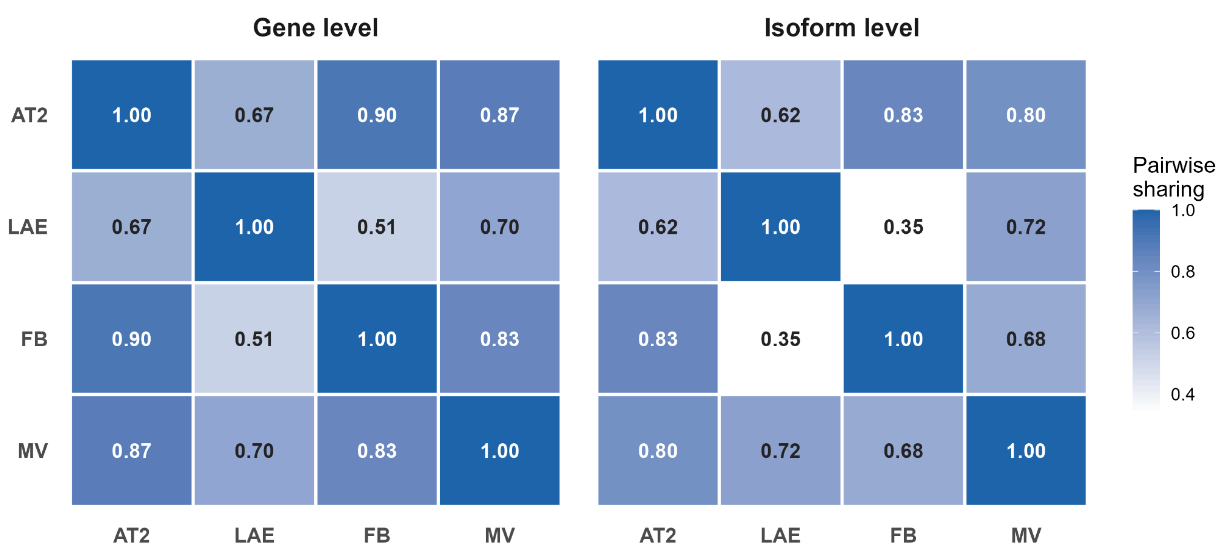
SF3 | Gene-Level Correlation of Short-read and Long-read Expression in All Cell Types. Hexbin density plots of short-read versus long-read gene expression (log2 CPM) for each cell type (AT2, LAE, FB, MV); color encodes bin count and the per-cell-type Spearman and Pearson correlations are annotated on each panel. **Abbreviations.** CPM, counts per million.



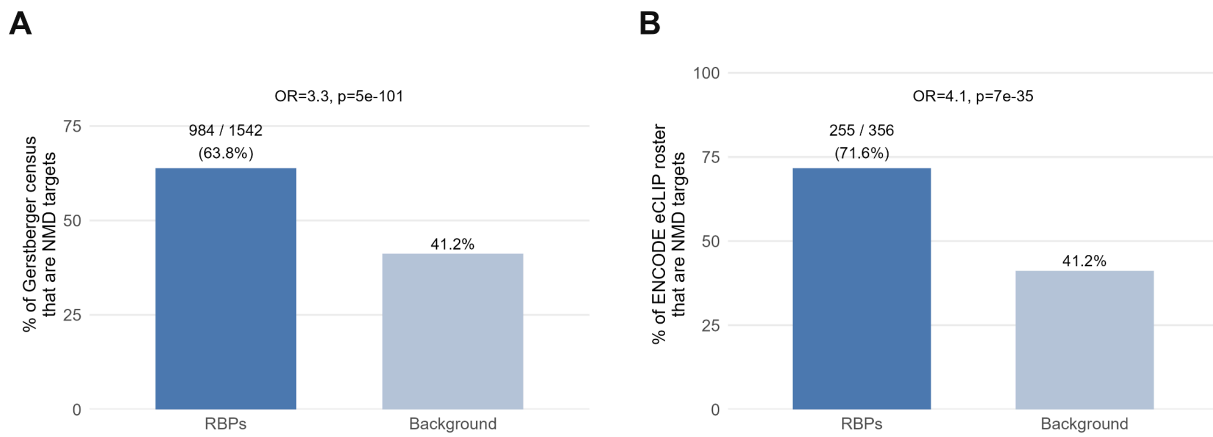
SF4 | Pairwise Expression Similarity and Isoform-Set Overlap Between Cell Types. A) Heatmap of pairwise Spearman correlations of mean per-isoform expression (log CPM) across the four cell types (AT2, LAE, FB, MV). **B)** Heatmap of pairwise Jaccard overlap of the expressed-isoform sets across the same cell types. **Abbreviations.** CPM, counts per million.



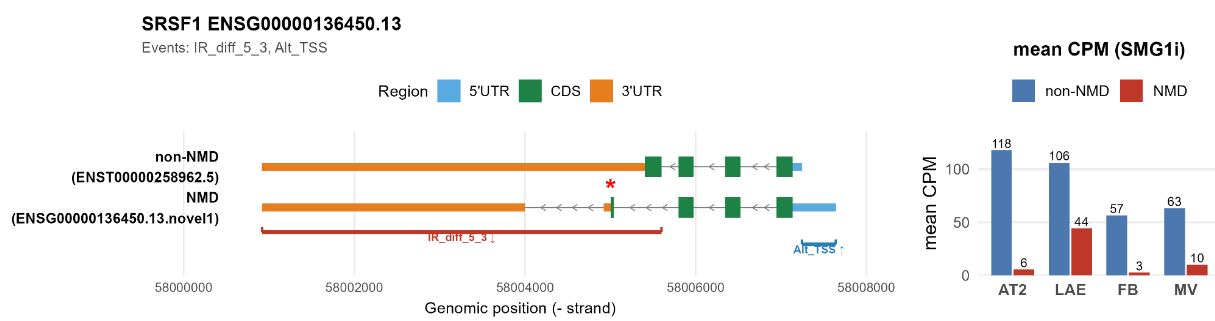
SF5 | Isoform Proportion versus Expression. A) All isoforms under DMSO and **B)** NMD susceptible isoforms, plotting each isoform's mean proportion of its gene's total expression against mean expression (log10 CPM), with reference lines at 0.10 and 0.50 proportion. **C)** NMD susceptible isoforms under DMSO versus SMG1i, with points colored by treatment. **Abbreviations.** CPM, counts per million.



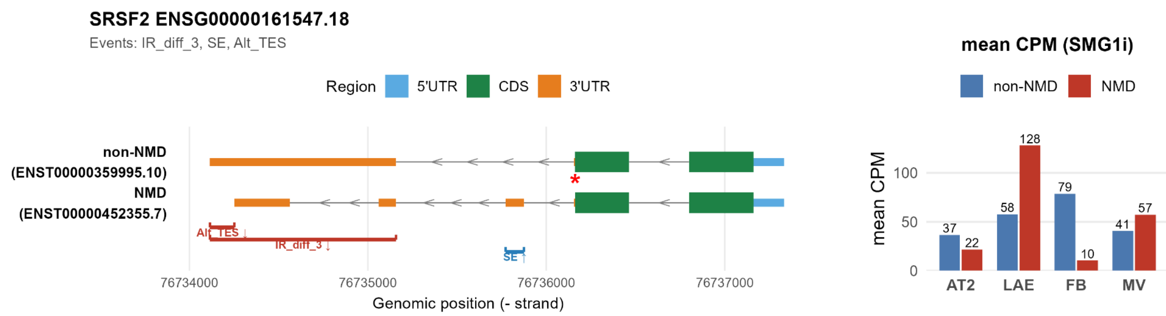
SF6 | Effect-Size Sharing at Gene and Isoform Resolution. Pairwise sharing heatmaps by mashr for short-read gene-level (left) and long-read isoform-level (right) NMD effects; each cell gives the fraction of features that share the same sign and lie within two logFC between the two cell types (AT2, LAE, FB, MV).



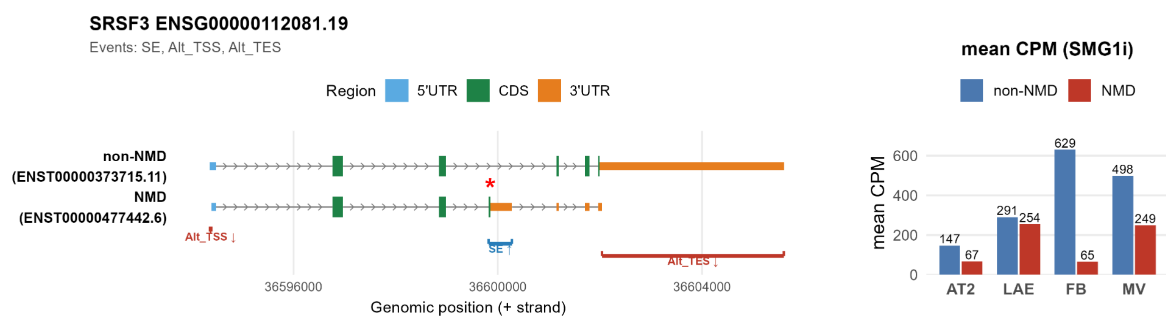
SF7 | RNA-Binding-Protein Enrichment among NMD Susceptible Genes. A) Fraction of the Gerstberger 2014 RBP census that are NMD targets versus the background rate. **B)** The same comparison against the ENCODE eCLIP roster (Van Nostrand 2020). An accompanying table reports enrichment for each RBP functional category (ST3). **Abbreviations.** RBP, RNA-binding protein; eCLIP, enhanced crosslinking immunoprecipitation.



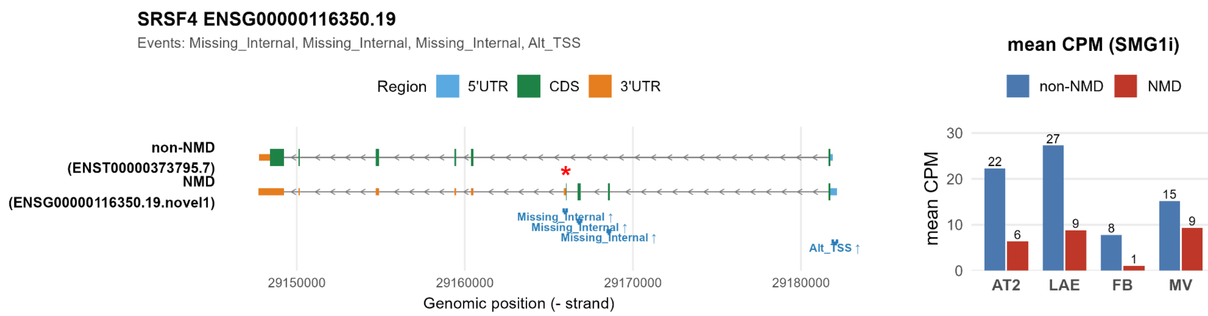
SF8 | SRSF1 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF1, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). **Abbreviations.** PTC, premature termination codon; CPM, counts per million; IR_diff 5'3', intron retention with both 5' and 3' boundary mismatches; Alt TSS, alternative transcription start site.



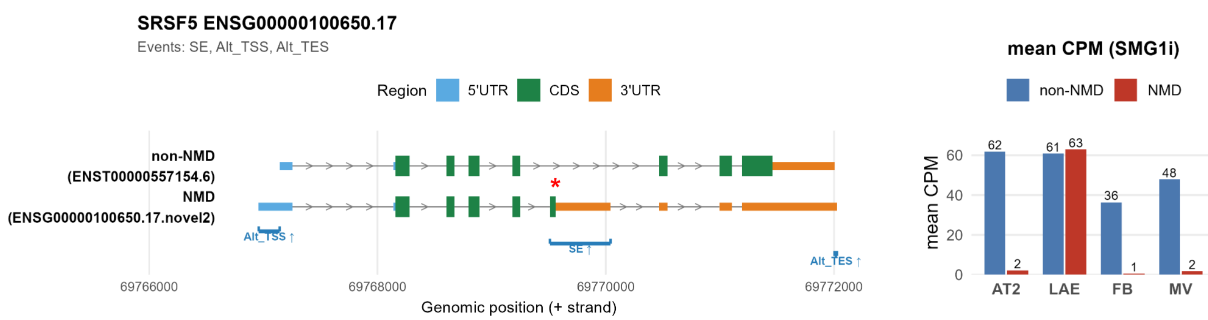
SF9 | SRSF2 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF2, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). **Abbreviations.** PTC, premature termination codon; CPM, counts per million; IR_diff 3', intron retention with a 3' boundary mismatch; SE, skipped exon; Alt TES, alternative transcription end site.



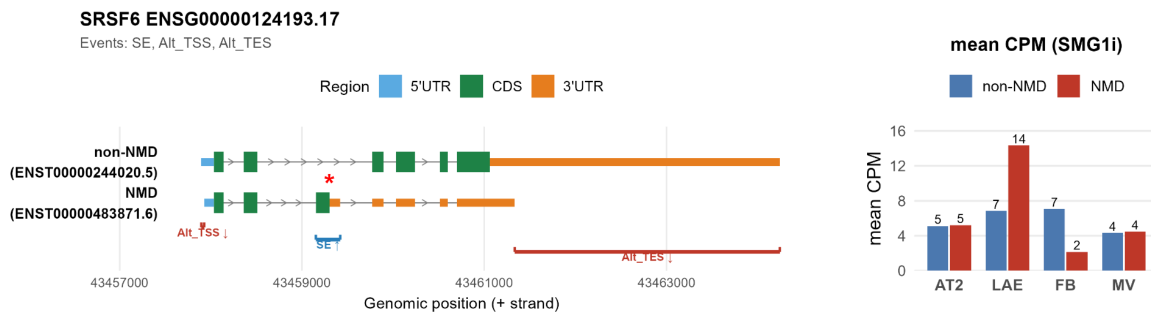
SF10 | SRSF3 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF3, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



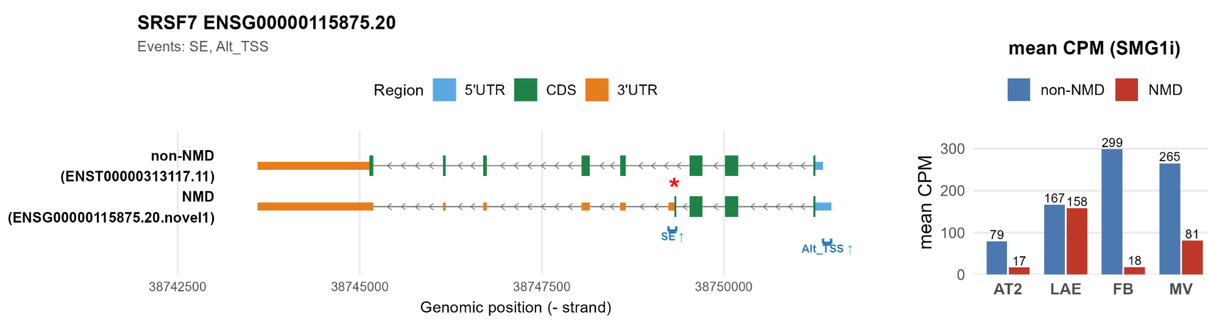
SF11 | SRSF4 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF4, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; Missing Internal, internal exon present in one isoform within the span of the other, outside strictly skipped exon criteria; Alt TSS, alternative transcription start site.



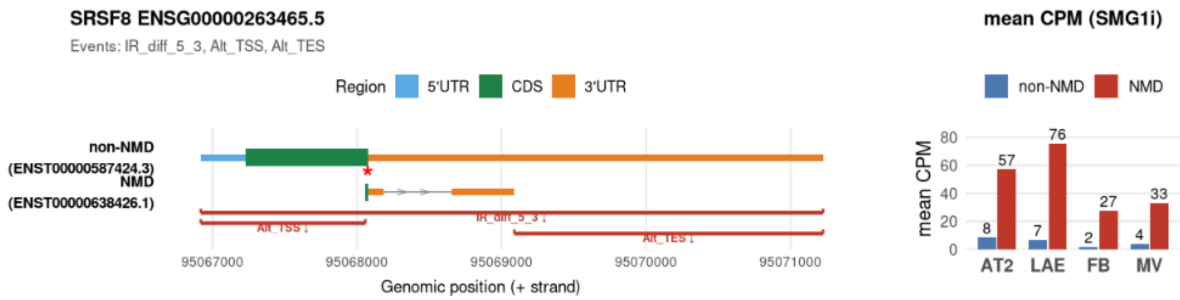
SF12 | SRSF5 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF5, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



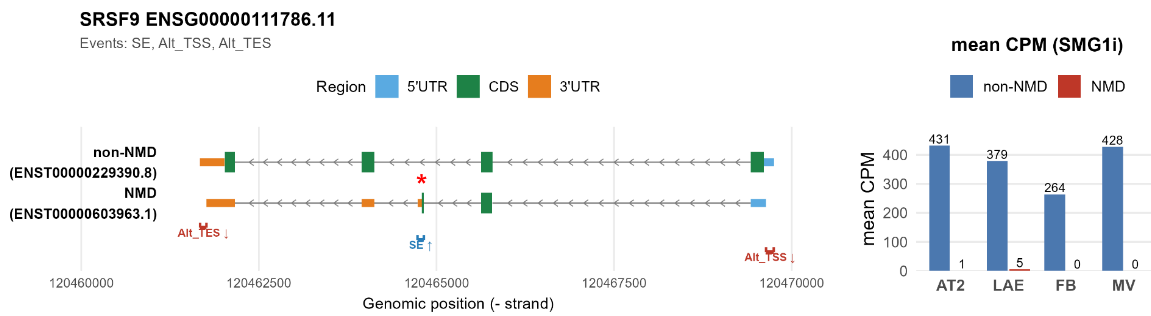
SF13 | SRSF6 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF6, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



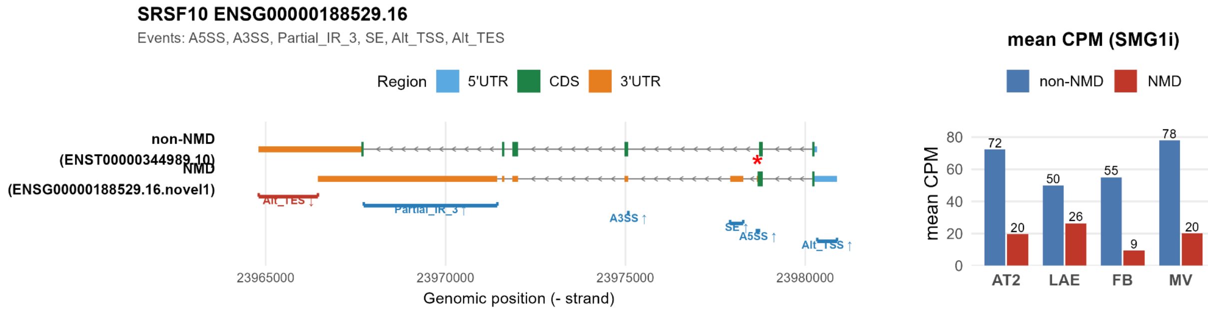
SF14 | SRSF7 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF7, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site.



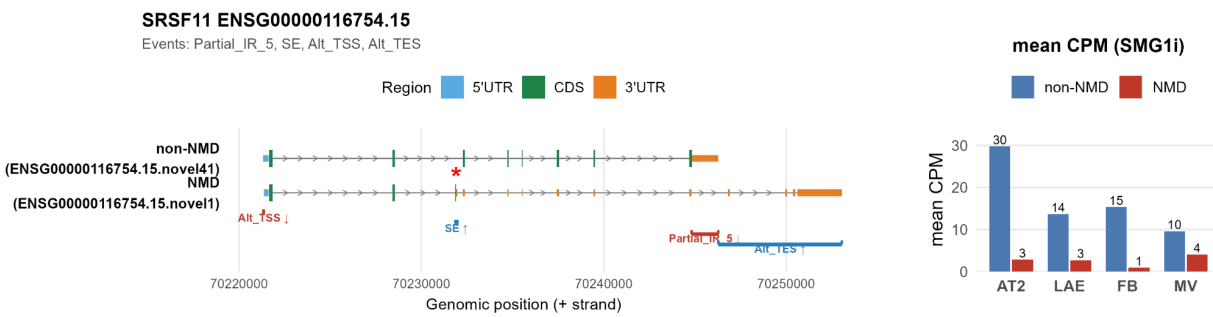
SF15 | SRSF8 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF8, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). **Abbreviations.** PTC, premature termination codon; CPM, counts per million; IR_diff 5'3', intron retention with both 5' and 3' boundary mismatches; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



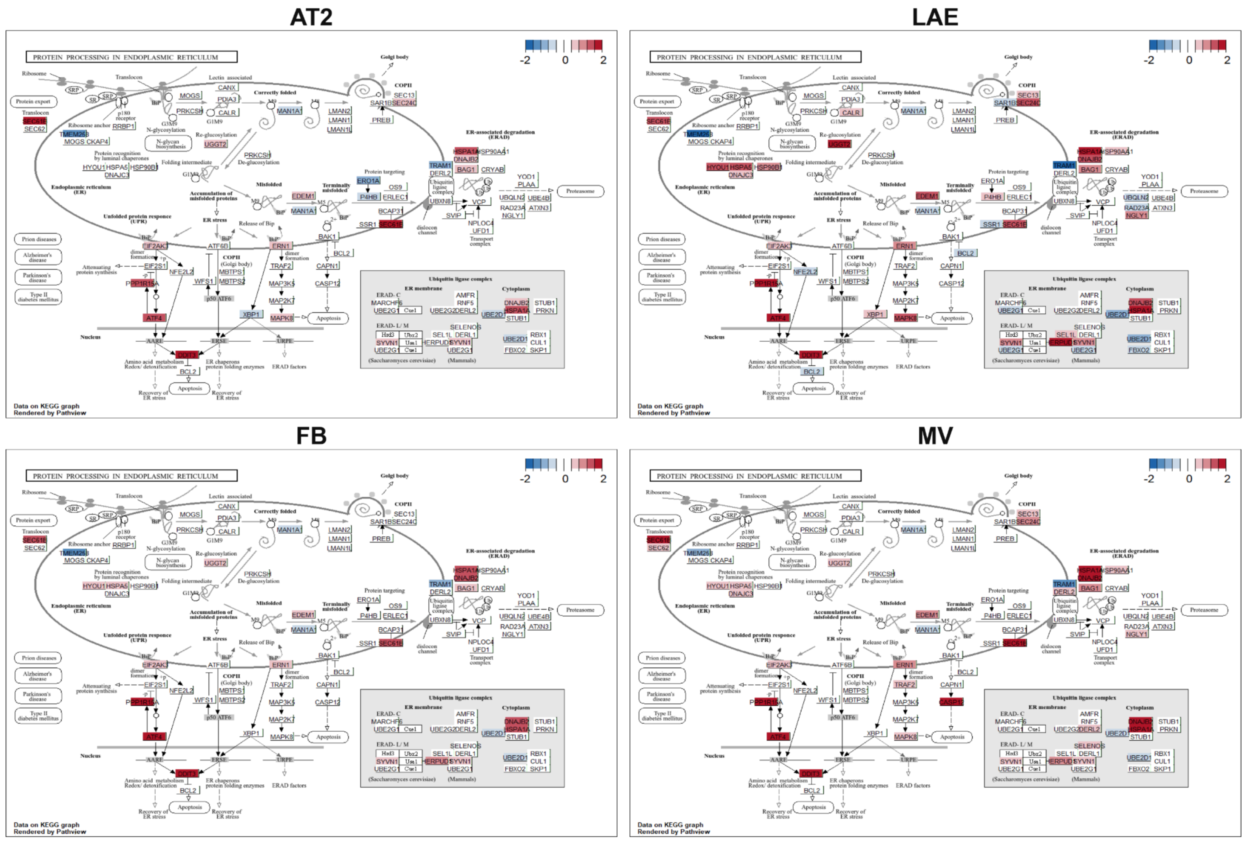
SF16 | SRSF9 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF9, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



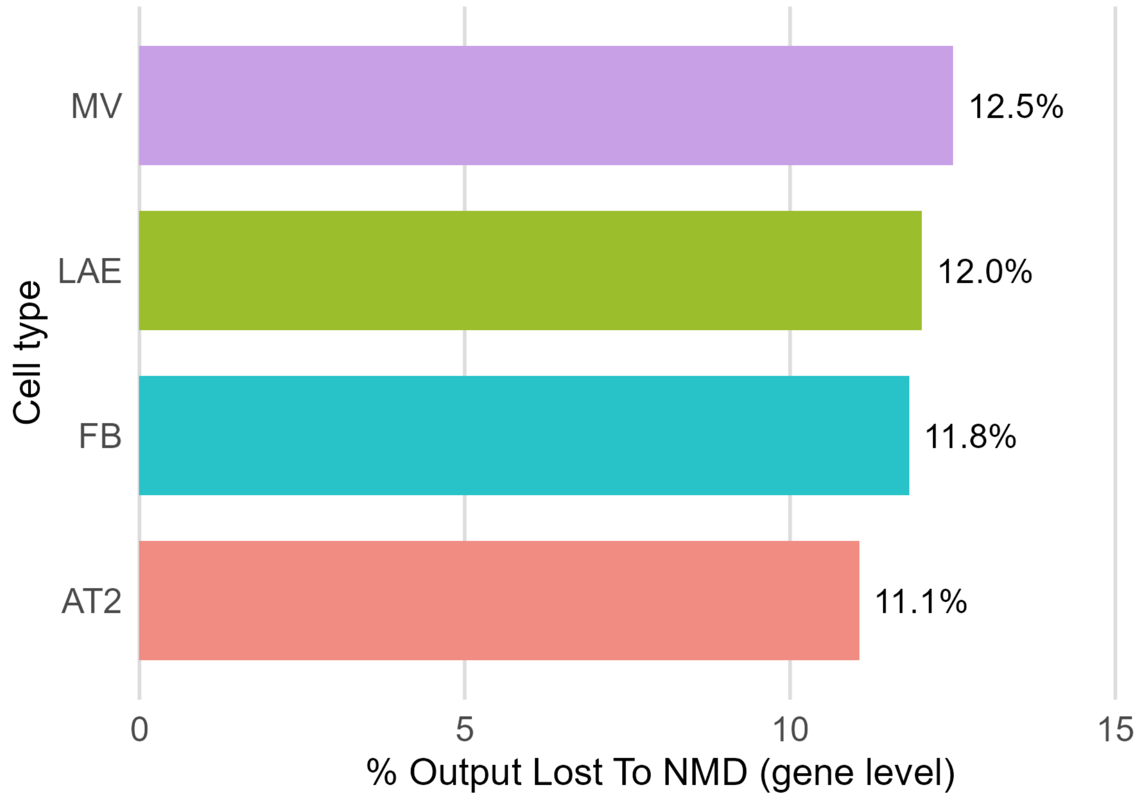
SF17 | SRSF10 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF10, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; A5SS, alternative 5′ splice site; A3SS, alternative 3′ splice site; Partial IR 3′, partial intron retention at the 3′ acceptor boundary; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



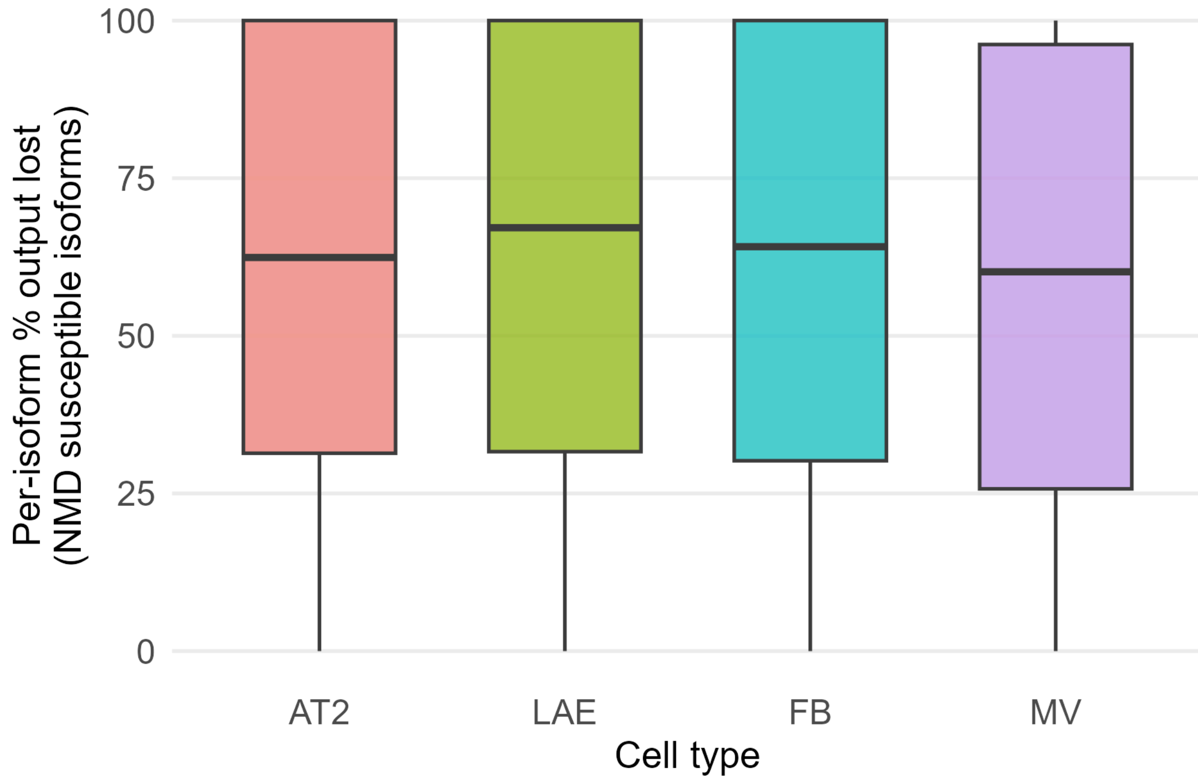
SF18 | SRSF11 Isoform Structure and Expression (non-NMD vs NMD). Exon/CDS structure of the most-expressed non-NMD reference isoform in DMSO and the top expressed NMD susceptible isoform in SMG1i of SRSF11, with splicing events annotated and the premature termination codon marked by a star. The side panel shows each isoform's mean CPM under SMG1i across the four cell types (AT2, LAE, FB, MV). Abbreviations. PTC, premature termination codon; CPM, counts per million; Partial IR 5′, partial intron retention at the 5′ donor boundary; SE, skipped exon; Alt TSS, alternative transcription start site; Alt TES, alternative transcription end site.



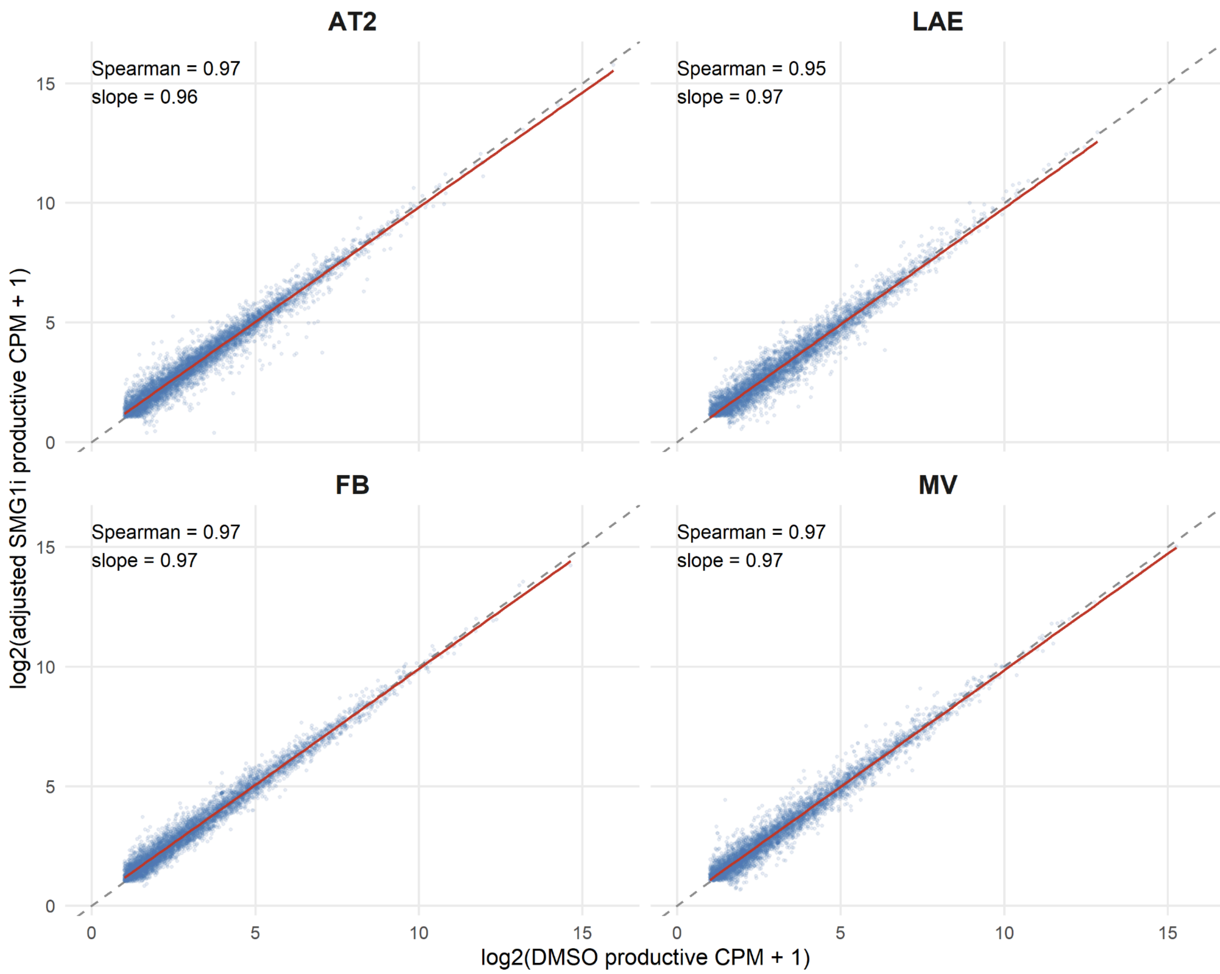
SF19 | Gene-Level NMD Susceptibility of the KEGG Protein Processing in Endoplasmic Reticulum Pathway in All Cell Types. The KEGG ER protein-processing pathway (hsa04141) shown for each cell type (AT2, LAE, FB, MV), with pathway genes colored by short-read NMD posterior log2 fold change. This pathway was selected because it contains ATF4, a key NMD-regulated transcription factor whose targets include multiple other pathway members, allowing visualization of how NMD susceptibility propagates both among co-regulated genes and to functionally downstream genes not themselves direct NMD substrates. Abbreviations. ER, endoplasmic reticulum; KEGG, Kyoto Encyclopedia of Genes and Genomes; NMD, nonsense-mediated decay.



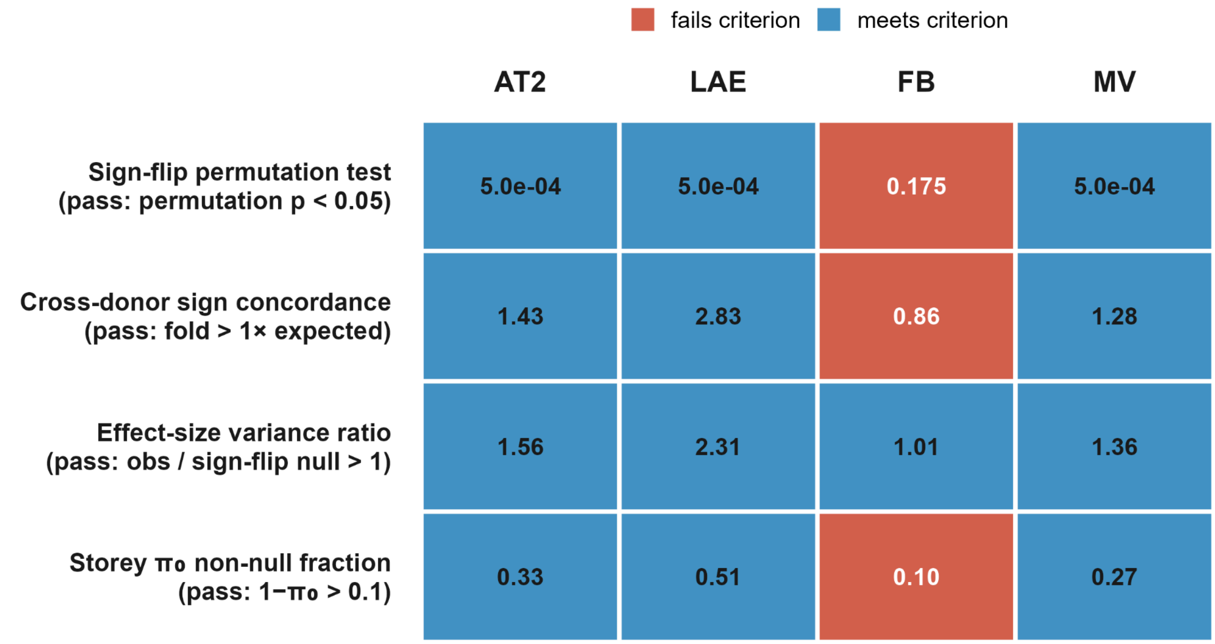
SF20 | Percent Transcriptional Output Lost to NMD, Gene Level. Per-cell-type percentage of gene-level transcriptional output attributable to NMD-susceptible isoforms (SMG1i minus DMSO), summarized across genes for each cell type (AT2, LAE, FB, MV).



SF21 | Per-Isoform Percent Output Lost. Distribution of the percentage of a gene's total output contributed by each individual NMD-susceptible isoform, shown per cell type (AT2, LAE, FB, MV).



SF22 | Correlation Between non-NMD Gene Expression in DMSO and Adjusted SMG1i. Per-cell-type scatter of non-NMD gene expression under DMSO versus SMG1i after the custom NMD-anchored normalization, by cell type (AT2, LAE, FB, MV).



SF23 | Reproducibility Tests of the Productive Response. Four reproducibility diagnostics, sign-flip permutation, cross-donor sign concordance, effect-size variance ratio, and Storey pi0 non-null fraction, shown per cell type (AT2, LAE, FB, MV) against the pass threshold indicated in each row label.